

University of Science and Technology of China

Topological Rigidity under Nonnegative Scalar Curvature

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Outline

1. Historical literature review
2. Main theorems and Motivation
3. Step 1 of the proof: topological reduction
4. Step 2 of the proof: from topology to a uniform flux gap
5. Step 3 of the proof: from the flux gap to contradiction
6. Step 4 of the proof: removing the nonparabolicity assumption

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Two rigidity questions

Wang's question

Let M^3 be contractible and admit a complete metric with

$$R_g \geq 0.$$

Must

$$M \cong_{\text{diff}} \mathbb{R}^3?$$

Gromov's question

Let M_γ be the interior of a compact handlebody of genus γ and admit a complete metric with

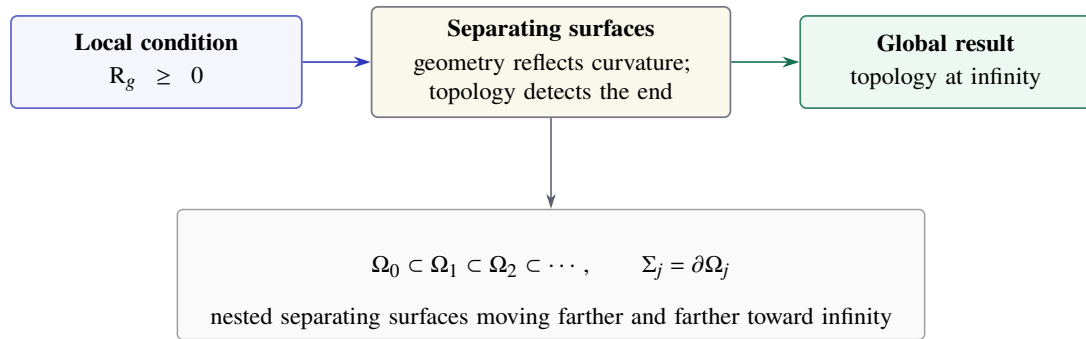
$$R_g \geq 0.$$

Must

$$\gamma \leq 1?$$

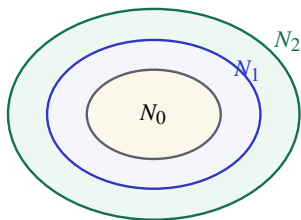
The difficulty is a mismatch of scales: **scalar curvature is local**, while the conclusion concerns **topology at infinity**.

The geometric problem



Recurring question: How can we construct separating surfaces farther and farther out whose **geometry is controlled by R_g** , while their **topology still detects the topology at infinity**?

Wang's genus-one setting [24,JDG]



$N_0 \subset N_1 \subset N_2 \subset \dots \nearrow M$
defining sequence of solid tori

Contractible genus-one manifolds

Wang considers

$$M = \bigcup_{k=0}^{\infty} N_k,$$

where each N_k is a solid torus and N_k is contractible in N_{k+1} .

But the nesting is nontrivial

Despite this homotopy triviality,

$$I(N_k, N_{k+1}) \geq 2.$$

$\pi_1(M) = 0$ does not see the topology at infinity; the **nesting of the solid tori** does.

Wang: topology at infinity versus the stable limit [24,JEMS]

Assume that M is contractible, $R_g > 0$, and $M \not\cong \mathbb{R}^3$.

Topology produces a persistent meridian

Choose handlebodies with Property (H) .

meridians on larger N_k

$\Downarrow$

area-minimizing disks

$\Downarrow$

$$\mathcal{L}_k \longrightarrow \mathcal{L} = \bigcup_{t \in \Lambda} L_t.$$

The topology forces a nontrivial meridian to **survive in the limiting lamination**.

$\pi_1^\infty = 0$ forces vanishing

Wang's Vanishing Property says that, for k large,

$$\gamma \subset L_t \cap \partial R_k \implies [\gamma] = 0 \in \pi_1(\partial R_k).$$

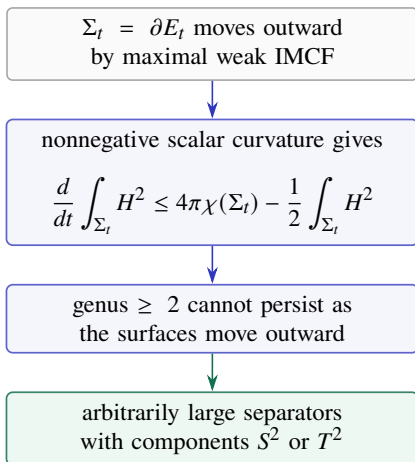
Failure of this property would produce a nontrivial element of $\pi_1^\infty(M)$.

Hence

$$\pi_1^\infty(M) = 0 \implies \mathcal{L} \text{ has the Vanishing Property.}$$

Topology forces a meridian to survive in $\mathcal{L}$, while $\pi_1^\infty(M) = 0$ forces it to vanish, contradiction.

Chodosh–Lai–Xu: the separator itself move outward [26,CPAM]



Green levels: analytic separating surfaces (CM [25,arXiv],YZ [26,arXiv])

Colding–Minicozzi: analytic control

For a positive Green function G , set

$$b = G^{-1}, \quad \Sigma_r = \{b = r\}.$$

Bochner–Gauss identities and Gauss–Bonnet relate the geometry and topology of Σ_r to differential inequalities for level-set energies.

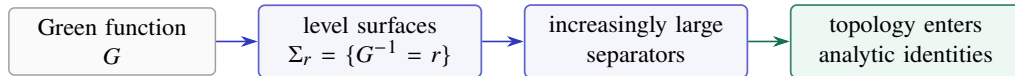
Yan–Zhu: increasingly large separators

If

$$G(x) \rightarrow 0 \quad (x \rightarrow \infty),$$

then $b = G^{-1}$ is proper.

Hence the level sets Σ_r move outward and exhaust the end as $r \rightarrow \infty$.



T. Colding–W. Minicozzi II, arXiv:2501.17947; Z. Yan–X. Zhu, arXiv:2604.08285.

The globalization bottleneck

One outward geometric family is asked to do several jobs

interact with R_g

remain compact
/ controlled

reach arbitrary
scales

detect topology
at infinity

Weak IMCF

Bounded geometry supplies the large-scale control needed to analyze proper, sweeping, and escaping behavior.

Global Green levels

The same Green levels must also exhaust the manifold. This requires decay/properness of G and large-scale gradient control, typically supplied by a Ricci lower bound.

Can Green-function levels keep their strong scalar-curvature calculus without also being responsible for exhausting the entire manifold?

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Main theorems

Contractible case

If (M^3, g) is complete, contractible, and $R_g \geq 0$, then

$$M \cong_{\text{diff}} \mathbb{R}^3.$$

Open-handlebody case

If the interior M_γ of a compact handlebody admits a complete metric with $R_g \geq 0$, then

$$\gamma \leq 1.$$

Yan–Zhu: how the global argument closes [26,arXiv]

Topology + CM calculus

If high genus persists along the Green levels, Gauss–Bonnet creates a definite gap in the Colding–Minicozzi inequalities:

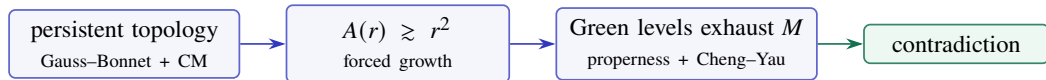
$$\text{persistent high genus} \implies A(r) \gtrsim r^2.$$

Thus topology forces increasingly strong **growth along the Green levels**.

Global control at infinity

In the nonparabolic setting, take a global minimal positive Green function G .

- Decay at infinity $G(x) \rightarrow 0$ gives $b = G^{-1}$ is proper, so $\Sigma_r = \{b = r\}$ themselves move to infinity.
- A Ricci lower bound gives Cheng–Yau gradient control, which is incompatible with the forced quadratic growth.



Yan–Zhu make **one global Green-level family do two jobs**: carry the CM analysis and simultaneously exhaust the manifold.

Motivation: keep the local engine, change the globalization

Colding–Minicozzi

Yan–Zhu

our paper

Common local engine: $G > 0 \rightarrow b = G^{-1} \rightarrow \Sigma_r = \{b = r\}$
Bochner–Gauss + Gauss–Bonnet $\Rightarrow$ topology forces Green-level growth

What we keep

The same local Green-level mechanism:

high genus $\Rightarrow$ Gauss–Bonnet gap $\Rightarrow A(r) \gtrsim r^2$.

This is the analytic engine of the argument.

What we change

We no longer require the Green levels themselves to exhaust the manifold.

Instead, we separate

r : local Green-level motion,
 τ : global exhaustion.

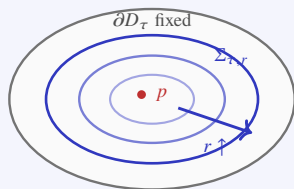
Decouple the two jobs: keep the Colding–Minicozzi analysis local, and introduce an independent outer exhaustion.

Turning point: decouple the two motions

For each exhaustion domain D_τ , let

$$G_\tau = G_{D_\tau}(p, \cdot), \quad b_\tau = G_\tau^{-1}, \quad \Sigma_{\tau,r} = \{b_\tau = r\}.$$

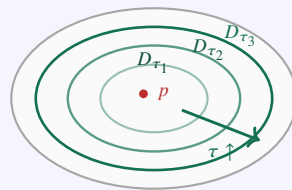
1. Fix τ : move the Green level



$$r \rightarrow \infty \implies \Sigma_{\tau,r} \rightarrow \partial D_\tau.$$

Here D_τ is fixed. This is the parameter used in the **Colding–Minicozzi level-set calculation**.

2. Increase τ : move the domain



$$D_{\tau_1} \subset D_{\tau_2} \subset \cdots, \quad D_\tau \nearrow M.$$

Here the **Dirichlet domain itself moves**. This is the parameter that carries the argument **globally toward infinity**.

Different choices of green function

Nonparabolicity does not give the properness we need

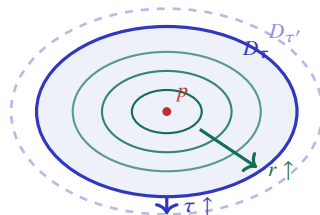
Nonparabolicity gives a minimal positive Green function G_M , but not necessarily $G_M(x) \rightarrow 0$ when $x \rightarrow \infty$. Hence G_M^{-1} need not be proper, so its levels may fail to detect the topology at infinity.

Localize

Choose an exhaustion $D_\tau \uparrow M$ and let G_τ be the Dirichlet Green function on D_τ . Then

$$G_\tau = 0 \text{ on } \partial D_\tau \implies b_\tau := G_\tau^{-1} \rightarrow \infty.$$

Thus the Green levels $\Sigma_{\tau,r} := \{b_\tau = r\}$ sweep from p toward ∂D_τ as $r \uparrow \infty$.



r : move Green levels

τ : enlarge the outer domain

Local Green functions provide the moving surfaces; the global Green function is used only for comparison.

Inner motion: From the CM growth to boundary flux

What the inner analysis gives

For every sufficiently large Green level in D_τ ,

$$A_\tau(r) \geq c_* r^2,$$

with $c_* > 0$ independent of τ .

Boundary limit

Set $q_\tau = -\partial_{\nu_\tau} G_\tau$. Then

$$\frac{A_\tau(r)}{r^2} \longrightarrow \int_{\partial D_\tau} q_\tau^2 dA \quad (r \rightarrow \infty).$$

Thus the quadratic CM growth becomes a **uniform positive boundary flux**.

$$\frac{A_\tau(r)}{r^2} \geq c_*$$

$r \rightarrow \infty$
levels approach ∂D_τ

$$\int_{\partial D_\tau} q_\tau^2 dA \geq c_*$$

Outer motion: boundary flux forces Robin growth

Now let the outer domain move

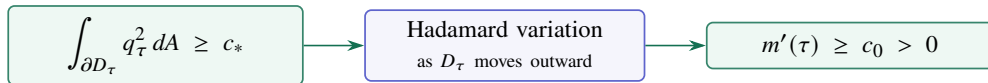
For each D_τ , write

$$m(\tau) := m_{D_\tau}(p)$$

for the **Robin constant** of the Dirichlet Green function G_τ .

Hadamard variation expresses its change in terms of the boundary flux:

$$m'(\tau) = \frac{1}{4\pi} \int_{\partial D_\tau} (\text{boundary speed}) q_\tau^2 dA.$$



The flux obtained from the inner r -analysis now becomes **uniform growth in the outer parameter τ** .

The contradiction

What the outer motion gives

From the uniform boundary flux,

$$m'(\tau) \geq c_0 > 0$$

on regular intervals.

Critical outer levels can only add **nonnegative jumps**.

Hence

$$m(\tau) \longrightarrow +\infty \quad (\tau \rightarrow \infty).$$

What nonparabolicity gives

If M is nonparabolic, there exists a **global minimal positive Green function**. Let $m_M(p)$ be its Robin constant.

Domain monotonicity gives

$$m(\tau) = m_{D_\tau}(p) \leq m_M(p) < \infty.$$

Therefore the previous growth is impossible.



The global Green function provides a finite Robin barrier. Its positive levels need not exhaust M .

From nonparabolic metric to general metric

Fixed-metric analytic theorem

For a complete, one-ended, **nonparabolic** 3-manifold,

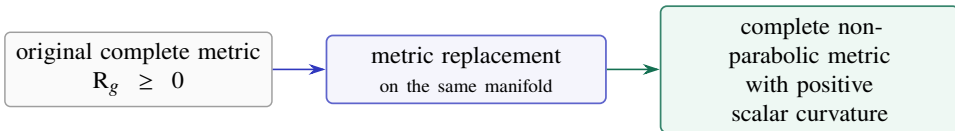
$$R_g \geq 0, \quad b_1(M) < \infty,$$

the Green-function argument forces low-genus separators at infinity.

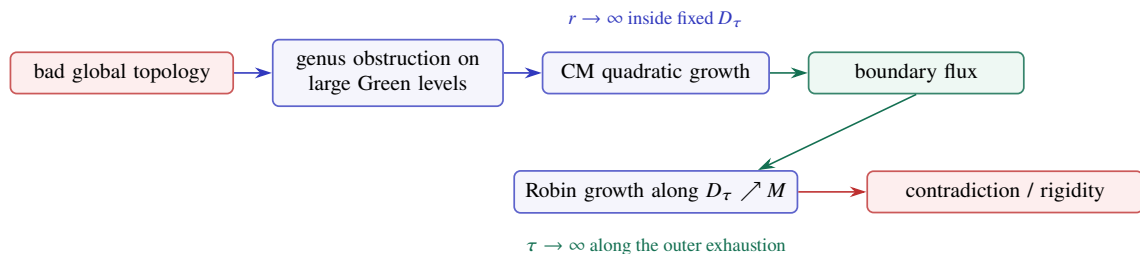
But our rigidity theorem assumes only

$$\text{completeness} \quad + \quad R_g \geq 0.$$

So one more step is needed: **remove the nonparabolicity assumption without changing the topology.**



Proof roadmap: two-stage transfer from topology to infinity



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The low-genus separator property

Admissible separator

A smooth precompact domain $U \Subset M$ such that

- U is connected;
- $M \setminus U$ is connected;
- ∂U is smooth, compact and connected.

Property (LG)

Every compact $K \Subset M$ is contained in an admissible separator U satisfying

$$\text{genus}(\partial U) \leq 1.$$

Property (LG) is the exact interface between the analytic argument and the topological rigidity theorem.

Why (LG) settles both rigidity questions

Contractible case

Repeatedly apply (LG) to obtain a nested exhaustion by connected surfaces of genus 0 or 1.

- sphere exhaustion $\Rightarrow M \cong \mathbb{R}^3$ (Husch–Price);
- torus exhaustion $\Rightarrow M \cong \mathbb{R}^3$ (Chodosh–Lai–Xu).

Open handlebody M_γ

A fixed compact core has end

$$\Sigma_\gamma \times (0, \infty).$$

Any admissible ∂U enclosing it maps to Σ_γ with degree ± 1 , so

$$\gamma \leq \text{genus}(\partial U) \leq 1.$$

From now on, geometry and analysis have one job: prove (LG).

Negating (LG): what this means for the Green levels

A fixed topological obstruction

Assume (LG) fails. Then there exists a compact set $K_0 \Subset M$ such that

$$K_0 \subset U, \quad U \text{ admissible} \implies \text{genus}(\partial U) \geq 2.$$

Apply the obstruction to Green levels

For each Dirichlet Green function G_τ , recall

$$b_\tau := G_\tau^{-1}, \quad U_{\tau,r} := \{b_\tau < r\}, \quad \Sigma_{\tau,r} := \{b_\tau = r\}.$$

For every regular value r , $\partial U_{\tau,r} = \Sigma_{\tau,r}$. Hence, once we prove $K_0 \subset U_{\tau,r}$ and $U_{\tau,r}$ is admissible, the contradiction hypothesis gives

$$\boxed{\text{genus}(\Sigma_{\tau,r}) \geq 2.}$$

The topological task is concrete: **make the Green superlevel domains contain K_0 and prove that their regular boundaries are admissible separators.**

Build a controlled outer exhaustion

To test the fixed obstruction K_0 with Green levels, we first need a nested family of bounded domains on which the Dirichlet Green functions can be compared.

Filled Morse exhaustion

Choose a proper Morse function

$$\rho : M \rightarrow [0, \infty), \quad |\nabla \rho| \leq L.$$

If C_τ is the component of $\{\rho < \tau\}$ containing p , fill all precompact complementary components:

$$D_\tau := C_\tau \cup \{\text{precompact complementary components}\}.$$

For regular τ , there holds $D_\tau \Subset M$, both D_τ and $M \setminus D_\tau$ are connected, and

$$D_\sigma \subset D_\tau \quad (\sigma < \tau), \quad D_\tau \uparrow M.$$

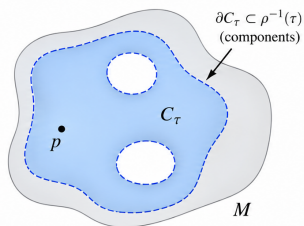
Moreover, the unit outward normal vector and the outward normal velocity satisfy

$$v_\tau = \frac{\nabla \rho}{|\nabla \rho|}, \quad V_\tau = \frac{1}{|\nabla \rho|} \geq L^{-1}.$$

The parameter τ gives a nested outer family with uniform control of its outward motion.

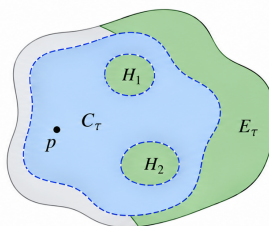
Construction of the filled exhaustion D_τ

(A) The sublevel set $\{\rho < \tau\}$
and its component C_τ



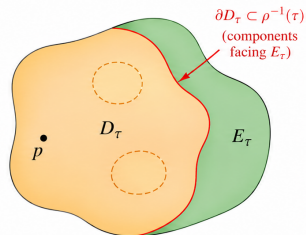
C_τ is the component of $\{\rho < \tau\}$ containing p .

(B) The complementary components
of $M \setminus C_\tau$



$$M \setminus C_\tau = E_\tau \sqcup H_1 \sqcup H_2$$

(C) The filled domain $D_\tau = M \setminus E_\tau$



$$D_\tau = C_\tau \cup H_1 \cup H_2 = M \setminus E_\tau$$

One threshold makes the Green levels admissible

Enlarge the fixed obstruction once and for all:

$$K \ni K_0 \cup S \cup \{p\},$$

where S is used to determine whether the boundary of a connected domain is connected. Since $D_\tau \uparrow M$, choose a regular τ_0 such that

$$K \Subset D_{\tau_0}.$$

A threshold independent of the outer domain

Since $G_{\tau_0} > 0$ on $K \setminus \{p\}$, choose $r_* > 0$ so that

$$G_{\tau_0} > r_*^{-1} \quad \text{on } K \setminus \{p\}.$$

Domain monotonicity of green functions gives

$$K \subset U_{\tau,r} \quad (\tau \geq \tau_0, r \geq r_*).$$

For regular $\tau \geq \tau_0$ and regular $r \geq r_*$, both $U_{\tau,r}$ and $M \setminus U_{\tau,r}$ are connected, and $S \subset U_{\tau,r}$ implies $\Sigma_{\tau,r}$ is connected. Hence $U_{\tau,r}$ is an admissible separator containing K_0 . By the failure of (LG),

$$\text{genus}(\Sigma_{\tau,r}) \geq 2.$$

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From the genus gap to a flux gap

The previous step produced constants τ_0 and r_* such that, for every sufficiently large regular τ and every regular $r \geq r_*$,

$$\Sigma_{\tau,r} \text{ is connected and } \text{genus}(\Sigma_{\tau,r}) \geq 2.$$

Fix such a τ . Write

$$G_\tau = G_{D_\tau}(p, \cdot), \quad b_\tau = G_\tau^{-1}, \quad q_\tau = -\partial_{\nu_\tau} G_\tau.$$

Our goal in this section is

$$\int_{\partial D_\tau} q_\tau^2 dA \geq c_* > 0,$$

where c_* is independent of the outer parameter τ .

$$\text{genus}(\Sigma_{\tau,r}) \geq 2 \implies \text{uniform boundary flux gap.}$$

Why quadratic growth is the right analytic target

Fix one sufficiently large regular τ and define

$$A_\tau(r) := r^{-2} \int_{\Sigma_{\tau,r}} |\nabla b_\tau|^2 dA.$$

Then

$$\frac{A_\tau(r)}{r^2} = \int_{\{G_\tau=r^{-1}\}} |\nabla G_\tau|^2 dA.$$

As $r \rightarrow \infty$, the Green level approaches ∂D_τ , hence

$$\lim_{r \rightarrow \infty} \frac{A_\tau(r)}{r^2} = \int_{\partial D_\tau} q_\tau^2 dA.$$

Therefore it is enough to prove $A_\tau(r) \geq c_* r^2$ with $c_* > 0$ independent of τ .

The CM identity: topology becomes positive forcing

Suppress τ for one fixed outer domain. The exact Colding–Minicozzi identity can be written as

$$rA'(r) = A(r) + \frac{1}{2r} \int_0^r (S_1(s) + B_2(s) - 2\kappa(s)) ds.$$

Here

$$S_1(r) = \int_{\Sigma_r} R_g dA \geq 0, \quad B_2(r) = \int_{\Sigma_r} \frac{|B|^2}{4r^2 |\nabla b|^2} dA_g \geq 0, \quad \kappa(r) = \int_{\Sigma_r} K_{\Sigma_r} dA.$$

For every regular $r \geq r_*$,

$$\text{genus}(\Sigma_r) \geq 2 \implies \kappa(r) = 2\pi\chi(\Sigma_r) \leq -4\pi,$$

so

$$-2\kappa(r) \geq 8\pi.$$

$$\text{genus gap} \xrightarrow{\text{Gauss–Bonnet}} \text{persistent positive forcing for } A.$$

Step 2.1: The genus gap enters the CM computational framework

Put $\beta := b_1(M)$. For every regular level, the component estimate and Gauss–Bonnet give

$$\kappa(r) = 4\pi \sum_{j=1}^{N(r)} (1 - g_j) \leq 4\pi(\beta + 1).$$

Combining this estimate with the estimate $\kappa(r) \leq -4\pi$ arising from the topological obstruction, we obtain that for every regular $r \geq r_*$,

$$\int_0^r (-2\kappa(s)) ds \geq -8\pi(\beta + 1)r_* + 8\pi(r - r_*) = 8\pi r - 8\pi(\beta + 2)r_*.$$

Hence

$$rA'(r) \geq A(r) + 4\pi - \frac{4\pi(\beta + 2)r_*}{r}.$$

Define $R_0 := 2(\beta + 2)r_*$, then, for $r \geq R_0$,

$$rA'(r) \geq A(r) + 2\pi.$$

With $r_1 := 2R_0$,

$$\left(\frac{A}{r}\right)' \geq \frac{2\pi}{r^2} \implies A(r_1) \geq 2\pi, \quad a(r_1) := \frac{r_1 A'(r_1)}{A(r_1)} \geq 1.$$

Step 2.2: derive a integral Riccati inequality

Define the logarithmic growth exponent

$$a(r) := \frac{rA'(r)}{A(r)}.$$

Recall $B = \text{Hess}(b^2) - 2|\nabla b|^2 g$, then B is trace-free and

$$\int_{\Sigma_r} B(\nu, \nu) dA = 2r^2 a(r) A(r).$$

Using Cauchy–Schwarz on Σ_r ,

$$\begin{aligned} B_2(r) &= \int_{\Sigma_r} \frac{|B|^2}{4r^2 |\nabla b|^2} dA_g \geq \frac{3}{8r^2} \int_{\Sigma_r} \frac{B(\nu, \nu)^2}{|\nabla b|^2} dA_g \\ &\geq \frac{3}{8r^2} \frac{\left(\int_{\Sigma_r} B(\nu, \nu) dA_g \right)^2}{\int_{\Sigma_r} |\nabla b|^2 dA_g} = \frac{3}{2} a(r)^2 A(r). \end{aligned}$$

The logarithmic growth rate admits such the control: $\frac{B_2(r)}{A(r)} \geq \frac{3}{2} a(r)^2$.

Step 2.2: derive an integral Riccati inequality

Set

$$W := |\nabla b^2|, \quad V := \frac{b^{-2}}{A(b)} \nabla W.$$

The weighted Bochner–Gauss computation gives, for a.e. regular $r \geq r_1$,

$$r \int_{\Sigma_r} \frac{\operatorname{div} V}{|\nabla b|} dA = \frac{S_1(r) + B_2(r) - 2\kappa(r)}{A(r)} + 2(1 - a(r)^2).$$

Using the geometric conditions together with the growth estimate for $a(r)$, we obtain

$$r \int_{\Sigma_r} \frac{\operatorname{div} V}{|\nabla b|} dA \geq 2 - \frac{1}{2}a(r)^2.$$

Moreover, $\partial_\nu W = \operatorname{Hess}(b^2)(\nu, \nu)$ implies

$$\int_{\Sigma_r} \langle V, \nu \rangle dA = 2a(r) + 2.$$

Thus, for regular $r_1 \leq s_1 < s_2$, Gauss–Green on $\{s_1 < b < s_2\}$ gives

$$a(s_2) - a(s_1) \geq \int_{s_1}^{s_2} \left(1 - \frac{a(r)^2}{4}\right) \frac{dr}{r}.$$

By continuity of a , the inequality extends to arbitrary $r_1 \leq s_1 < s_2$.

Step 2.3: Riccati comparison gives the quadratic growth

The comparison equation is

$$r\varphi'(r) = 1 - \frac{\varphi(r)^2}{4}, \quad \varphi(r_1) = 1.$$

Since $a(r_1) \geq 1$, the integral comparison principle gives

$$a(r) \geq \varphi(r) := 2 \frac{3r - r_1}{3r + r_1} \quad (r \geq r_1).$$

Now integrate the logarithmic growth:

$$\log \frac{A(r)}{A(r_1)} = \int_{r_1}^r a(s) \frac{ds}{s} \geq 2 \log \frac{r}{r_1} - 4 \log \frac{4r}{3r + r_1}.$$

Thus we obtain the growth estimate for $A(r)$. Combining this with $A(r_1) \geq 2\pi$ and $(3r + r_1)/(4r) \geq 3/4$,

$$A(r) \geq c_* r^2, \quad c_* := \frac{2\pi}{r_1^2} \left(\frac{3}{4}\right)^4 = \frac{81\pi}{128r_1^2}.$$

The constants r_1 and c_* depend only on $b_1(M)$ and r_* , hence are independent of τ .

Step 2.4: returns to the boundary

Restore the outer-domain subscript. For every sufficiently large regular τ ,

$$A_\tau(r) \geq c_* r^2 \quad (r \geq r_1),$$

with the same constant $c_* > 0$.

On $\Sigma_{\tau,r} = \{b_\tau = r\} = \{G_\tau = r^{-1}\}$, we have

$$\frac{A_\tau(r)}{r^2} = \int_{\{G_\tau = r^{-1}\}} |\nabla G_\tau|^2 dA.$$

Boundary regularity gives

$$\int_{\partial D_\tau} q_\tau^2 dA = \lim_{r \rightarrow \infty} \frac{A_\tau(r)}{r^2} \geq c_*.$$

genus gap $\implies$ quadratic growth $\implies$ uniform boundary flux gap.

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The outer parameter now closes the proof

Return to Hadamard's formula:

$$\frac{d}{d\tau} m_{D_\tau}(p) \geq \frac{1}{4\pi L} \int_{\partial D_\tau} q_\tau^2 dA \geq \frac{c_*}{4\pi L}$$

on every regular interval. Critical Morse values contribute only nonnegative jumps.

Therefore

$$m_{D_\tau}(p) \longrightarrow +\infty \quad (\tau \rightarrow \infty).$$

But the global minimal Green function gives

$$m_{D_\tau}(p) \leq m_M(p) < \infty.$$

Contradiction.

Hence (LG) holds. The main Green-function argument is finished.

The fixed-metric theorem

Nonparabolic low-genus separators

Let (M^3, g) be complete, orientable, one-ended, and nonparabolic. If

$$R_g \geq 0, \quad b_1(M) < \infty,$$

then M satisfies (LG).

Where topology enters

One-endedness and finite b_1 force sufficiently large Green levels to be admissible; failure of (LG) then forces their genus to be at least two.

Where nonparabolicity enters

Only at the final line:

$$m_{D_\tau}(p) \leq m_M(p) < \infty.$$

No behavior of G_M at infinity is used.

Outline

1. Historical literature review
2. Main theorems and Motivation
3. Step 1 of the proof: topological reduction
4. Step 2 of the proof: from topology to a uniform flux gap
5. Step 3 of the proof: from the flux gap to contradiction
- 6. Step 4 of the proof: removing the nonparabolicity assumption**

The remaining task

The fixed-metric theorem applies to a complete, one-ended, *nonparabolic* metric with nonnegative scalar curvature. But the original rigidity statements assume only

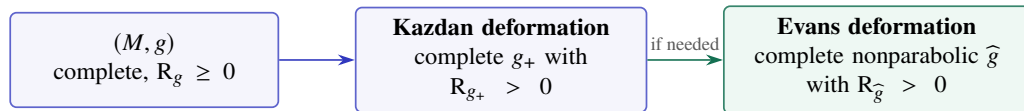
$$R_g \geq 0,$$

with no nonparabolicity hypothesis.

Can we replace the metric, without changing the manifold, so that the fixed-metric theorem becomes applicable?

The nonflat case: two deformations with separate jobs

Assume $\text{Ric}_g \not\equiv 0$. We use two metric deformations with different purposes:



Kazdan's job

Create a *strict* scalar-curvature:

$$R_{g_+} > 0.$$

Evans' job

If necessary, deform the metric at infinity so that it becomes *nonparabolic*.

Kazdan deformation: make the scalar curvature positive

Assume $R_g \geq 0$, $\text{Ric}_g \not\equiv 0$. Recall the Rayleigh quotient

$$\mu_1(\Omega, h) = \inf_{0 \neq v \in H^1(\Omega)} \frac{\int_{\Omega} (c_n |\nabla v|_h^2 + R_h v^2) dV_h}{\int_{\Omega} v^2 dV_h}. \quad (1)$$

Step 1. Produce a positive eigenvalue

For $\mathcal{L}_h := -8\Delta_h + R_h$, find $g_0 \sim g$ and $\Omega \Subset M$ such that

$$\mu_1(\Omega, g_0) > 0.$$

If $R_g \not\equiv 0$, take $g_0 = g$; if $R_g \equiv 0$, apply a compactly supported perturbation to the metric: $g_t = g - t\eta \text{Ric}_g$, then

$$\mu'_1(0) = \frac{1}{\text{Vol}_g(\Omega)} \int_{\Omega} \eta |\text{Ric}_g|^2 dV_g > 0.$$

Step 2. Kazdan's criterion's output

Kazdan's criterion gives a bounded positive function w with $0 < c_0 \leq w \leq C_0$ and $\mathcal{L}_{g_0} w > 0$ on M .

Kazdan deformation: make the scalar curvature positive.

Step 3. Conformal deformation

Set

$$g_+ = w^4 g_0.$$

Then

$$R_{g_+} = w^{-5} \mathcal{L}_{g_0} w > 0.$$

Since $g_0 \sim g$ and w is bounded above and below,

$$g_+ \sim g \implies g_+ \text{ is complete.}$$

Why do we need this deformation?

The next Evans deformation may lower scalar curvature on a compact transition region K .

With only $R_g \geq 0$, there is no room for such a loss; but now

$$R_{g_+} > 0 \implies \min_K R_{g_+} > 0.$$

This positive margin can absorb the error introduced by the Evans deformation.

Evans deformation

Now suppose (M, g_+) is still parabolic. One-endedness gives a proper exterior Evans potential

$$\Delta_{g_+} u = 0, \quad u|_{\partial E} = 0, \quad u(x) \longrightarrow \infty \quad \text{at infinity.}$$

Choose regular values $0 < a < b$ and a smooth nondecreasing function

$$\eta : [0, \infty) \rightarrow [0, 1], \quad \eta = 0 \text{ on } [0, a], \quad \eta = 1 \text{ on } [b, \infty).$$

Set

$$F_\varepsilon(t) = 1 + \varepsilon \int_0^t \eta(s) ds,$$

Define the global conformal factor

$$\phi = \begin{cases} F_\varepsilon(u), & \text{on } E, \\ 1, & \text{on } M \setminus E, \end{cases} \quad \boxed{\widehat{g} = \phi^4 g_+}.$$

Why capacity? Positive capacity of a compact set is equivalent to nonparabolicity (Grigor'yan). The linear growth of F_ε at infinity is chosen precisely to produce such positive capacity.

Evans deformation: preserve positivity and completeness

Scalar curvature

$$R_{\widehat{g}} = \phi^{-5}(-8\Delta_{g_+}\phi + R_{g_+}\phi).$$

Since

$$\Delta_{g_+}\phi = F''_{\varepsilon}(u)|\nabla u|^2,$$

the extra term is supported only on the compact band $\{a \leq u \leq b\}$. For ε small, it is dominated by $R_{g_+} > 0$.

Completeness

Since

$$\phi \geq 1,$$

we have

$$\widehat{g} = \phi^4 g_+ \geq g_+.$$

Thus divergent curves cannot become shorter, so $\widehat{g}$ remains complete.

For sufficiently small ε , $\widehat{g}$ is complete, $R_{\widehat{g}} > 0$.

Evans deformation: create nonparabolicity

Target

To prove $(M, \widehat{g})$ is nonparabolic, it is enough to show that some compact set has positive capacity. For a regular value $T > b$, define

$$I_T := \int_b^T F_\varepsilon(s)^{-2} ds, \quad h_T(x) := \frac{1}{I_T} \int_{u(x)}^T F_\varepsilon(s)^{-2} ds;$$
$$K_b := (M \setminus E) \cup \{u \leq b\}, \quad \Omega_T := (M \setminus E) \cup \{u < T\},$$

It suffices to prove

$$\lim_{T \rightarrow \infty} \text{Cap}_{\widehat{g}}(K_b; \Omega_T) = \lim_{T \rightarrow \infty} \int_{\{b < u < T\}} |\nabla h_T|_{\widehat{g}}^2 dV_{\widehat{g}} > 0.$$

Why choose h_T this way?

We seek h_T in the form $h = H(u)$ as a suitable representative for computing the capacity. The equation $\Delta_{\widehat{g}} h = 0$ reduces to $(F_\varepsilon^2 H')' = 0$, hence

$$H'(t) \propto F_\varepsilon(t)^{-2}.$$

Imposing $H(b) = 1, H(T) = 0$, gives the form used above.

Evans deformation: create nonparabolicity

Then the coarea formula gives

$$\text{Cap}_{\widehat{g}}(K_b; \Omega_T) = \int_{\{b < u < T\}} |\nabla h_T|_{\widehat{g}}^2 dV_{\widehat{g}} = \text{const} \int_b^T F_{\varepsilon}(t)^2 H'_T(t)^2 dt = \boxed{\frac{\text{const}}{I_T}}.$$

Finish the verification

For $t \geq b$, $F_{\varepsilon}(t) = F_{\varepsilon}(b) + \varepsilon(t - b)$, so

$$I_T \longrightarrow \int_b^{\infty} F_{\varepsilon}(t)^{-2} dt = \frac{1}{\varepsilon F_{\varepsilon}(b)} < \infty.$$

Therefore

$$\boxed{\text{Cap}_{\widehat{g}}(K_b) = \frac{J}{I_{\infty}} = J\varepsilon F_{\varepsilon}(b) > 0} \quad \implies \quad \boxed{(M, \widehat{g}) \text{ is nonparabolic.}}$$

Now the fixed-metric theorem applies

For the nonflat case we have constructed, on the *same manifold* M , a metric $\widehat{g}$ satisfying

$$\widehat{g} \text{ complete}, \quad R_{\widehat{g}} > 0, \quad (M, \widehat{g}) \text{ nonparabolic.}$$

The hypotheses

$$\text{orientable}, \quad \text{one-ended}, \quad b_1(M) < \infty$$

are topological and therefore unchanged.

The fixed-metric theorem applied to $(M, \widehat{g})$ gives

M satisfies (LG).

The conclusion concerns the manifold, not the chosen metric.

The flat case is separate and elementary

If $\text{Ric}_g \equiv 0$, then in dimension three $\text{Rm}_g \equiv 0$, so g is flat.

Contractible manifold

Complete + simply connected + flat gives

$$(M, g) \cong (\mathbb{R}^3, g_{\text{eucl}}).$$

Hence

$$M \cong_{\text{diff}} \mathbb{R}^3.$$

Open handlebody

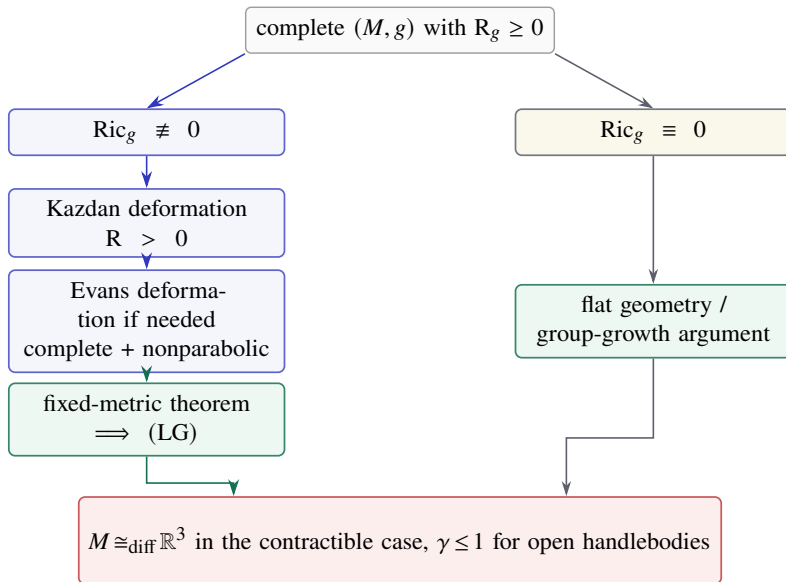
For M_γ ,

$$\pi_1(M_\gamma) \cong F_\gamma.$$

By the Euclidean packing lemma, the fundamental group of a complete flat 3-manifold, if finitely generated, has polynomial growth (of degree at most 3), whereas F_γ has exponential growth for $\gamma \geq 2$. Therefore

$$\gamma \leq 1.$$

Putting Step 4 together



Thank you

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